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Faculty of
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SPECIAL TOPIC IN DATA ENGINEERING (SECP3843)

SECTION 01

TUTORIAL ARTICLE

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1.1 Main Problem

Top management has to make decisions without real-time and reliable performance data of core business processes (sales, production) due to the lack of proper measurement and monitoring system in MPM, which diminishes their competitiveness.

1.2 Main Solution

A Business Intelligence (BI) dashboard system with Google Sheets data warehouse, using the Waterfall methodology, consisting of two dashboards: Sales Performance Dashboard and Production Performance Dashboard.

1.3 User Story

In Table 1.3.1, it describes the user story for viewing the sales performance dashboard, which allows top management to track revenue, compare forecast sales to actual sales and view product breakdowns to see if targets are being achieved.

User story: View sales performance dashboard
ID: US001
User Story Description As a member of top management at PT. MPM I want to view the sales performance dashboard showing revenue, forecast vs actual sales, and product category breakdowns so that I can assess whether sales operations are meeting KPI targets and make informed strategic decisions.
Flow of events: 1. The user logs in to the BI dashboard system. 2. The system displays the sales performance dashboard as the default landing view. 3. The user selects a date range, quarter, customer ID, or product category from the control chart. 4. The system filters and refreshes all charts to reflect the selected parameters. 5. The user reviews pie charts showing customer and product performance by volume and revenue. 6. The user reviews the heatmap table comparing forecast versus actual values by month. 7. The user reviews the combo chart to contrast multiple datasets simultaneously.
Alternative flow 1: 1. If no data exists for the selected date range, the system displays a 'no data available' message and prompts the user to adjust the filter.

Acceptance Criteria 1. The dashboard displays revenue, average revenue, and total sales scorecards correctly. 2. The heatmap table renders colour-coded cells reflecting performance ranking. 3. All charts update when filter parameters are changed. 4. Forecast vs actual values are visually distinguishable in the combo chart.
Exception flow 1: 1. If the user's session expires due to inactivity, the system automatically logs the user out and redirects to the login page.

Table 1.3.1 User story: View sales performance dashboard

Table 1.3.2 displays the user story about the production performance dashboard, describing how operational managers and top management monitor Overall Equipment Effectiveness (OEE), as well as planned and actual counts and specific machine metrics.

User story: View production performance dashboard
ID: US002
User Story Description As a member of top management or an operational manager I want to view the production performance dashboard showing OEE, planned vs actual production counts, and machine-level quality metrics so that I can evaluate productivity and identify underperforming machines or production lines.
Flow of events: 8. The user logs in and navigates to the production performance dashboard. 9. The user selects a date range, product division, or specific machine from the control chart. 10. The system displays gauge charts showing Performance, Availability, and Quality ratios. 11. The user views the bar chart comparing planned versus actual production counts. 12. The user reviews the OEE pivot heatmap table for each machine per month. 13. The user expands or collapses rows using the expand-collapse function.
Alternative flow 1: 2. If a specific machine has no recorded data for the selected period, its row in the heatmap is greyed out with a 'no data' indicator.

Acceptance Criteria - OEE gauge charts correctly display Performance, Availability, and Quality as percentages. - Planned and actual production scorecard values match the data warehouse figures. - Heatmap cell colours correctly encode machine rank (darker = higher rank). - The expand-collapse pivot table function operates without error.
Exception flow 1: If the data warehouse connection is unavailable, the system displays an error message and advises the data analyst to check the Google Sheets data source.

Table 1.3.2 User story: View production performance dashboard

In Table 1.3.3, it displays the user story for Uploading and Updating Warehouse Data, which describes the process of a data analyst cleaning and transforming raw ERP reports into the data warehouse in Google Sheets.

User story: Upload and update warehouse data
ID: US003
User Story Description As a data analyst I want to upload new sales and production report data into the Google Sheets data warehouse so that the BI dashboards always reflect the most current and accurate company performance data.
Flow of events: - The data analyst receives raw sales and/or production reports from the ERP system and relevant departments. - The analyst converts all raw data into Google Sheets format. - The analyst performs data cleaning — removing extraneous fields and correcting data types (e.g. converting string dates to date format). - The analyst runs the ETL transformation process to restructure the data into the warehouse schema. - The transformed data is loaded into the respective sales or production data warehouse sheet. - The analyst verifies the updated data is correctly reflected in the BI dashboard.
Alternative flow 1: If data already exists in Google Sheets format, the analyst skips the conversion step and proceeds directly to data cleaning.

Acceptance Criteria 9. Uploaded data is correctly structured according to the sales or production data warehouse schema. 10. Date fields are stored in the correct date format (not as strings). 11. The BI dashboard reflects newly uploaded data without requiring a manual rebuild. 12. No duplicate or extraneous records are present after the ETL process.
Exception flow 1: 3. If the uploaded data contains formatting errors or missing required fields, the system flags the affected rows and the analyst must correct and re-upload before the dashboard updates.

Table 1.3.3 User story: Upload and update warehouse data

In Table 1.3.4, it presents the user story: Filter and drill down into dashboard data, which describes how users can use parameters such as date ranges or product divisions to view performance metrics at a specific level of granularity.

User story: Filter and drill-down on dashboard data
ID: US004
User Story Description As a top management member or operational manager I want to apply filters (date range, quarter, customer, product category, machine, product division) and drill down into specific data segments so that I can analyse performance at a granular level and identify specific issues or trends.
Flow of events: 20. The user is viewing the sales or production performance dashboard. 21. The user selects one or more filter options from the control chart at the top of the dashboard. 22. The system dynamically updates all charts, tables, and scorecards to reflect the filtered data. 23. The user optionally uses the drill-down analysis feature to explore a specific metric in more detail. 24. The user uses the optional metric selector to switch between available KPI views.
Alternative flow 1: 4. If the user clears all filters, the dashboard reverts to displaying the full unfiltered dataset.

Acceptance Criteria 13. All filter controls (date, quarter, customer, product category, machine, division) function correctly and update all dashboard components. 14. Drill-down and optional metric features are accessible and return relevant data subsets. 15. Multiple filters can be applied simultaneously without error.
Exception flow 1: 4. If the combination of selected filters returns zero records, the system displays a clear 'no matching data' message instead of blank or broken charts.

Table 1.3.4 User story: Filter and drill-down on dashboard data

Table 1.3.5 explains the user story for 'Share and Export Dashboard Reports' in Table 1.3.5 and how users can create shareable links or print files in Google Data Studio to provide insights to others.

User story: Share and export dashboard report
ID: US005
User Story Description As a top management member or operational manager I want to share the dashboard link across platforms or print/export the report so that I can distribute performance insights to relevant stakeholders who may not have direct dashboard access.
Flow of events: 25. The user is viewing a dashboard with the desired filters already applied. 26. The user clicks the share feature in Google Data Studio. 27. The system generates a shareable link or sharing options across platforms. 28. Alternatively, the user selects the print/export option to generate a printable report. 29. The system generates the report in the desired format and makes it available for download or printing.
Alternative flow 1: 5. If the recipient does not have a Google account, the user sets dashboard access to 'Anyone with the link' to allow viewing without authentication.

Acceptance Criteria 16. A valid shareable link is generated and the shared dashboard is viewable by the intended recipients. 17. The printed or exported report accurately reflects the current dashboard state including applied filters. 18. The share feature works across platforms as described in Google Data Studio functionality.
Exception flow 1: 5. If the user does not have sharing permissions, the system displays an access denied message and prompts the user to contact the data analyst or system administrator.

Table 1.3.5 User story: Share and export dashboard report

It is defined in Table 1.3.6 as the user story responsible for providing the ability for an authorized data analyst or manager to change, move or update a visual element to suit changing business requirements.

User story: Manage dashboard layout
ID: US006
User Story Description As a data analyst or operational manager I want to edit the dashboard layout — adding, removing, or repositioning visual elements so that the dashboard continues to meet evolving business requirements and management preferences.
Flow of events: 30. The authorised user opens the dashboard in edit mode in Google Data Studio. 31. The user adds, removes, or repositions charts, scorecards, or tables as required. 32. The user adjusts visualisation settings such as chart type or displayed metrics. 33. The user saves the updated dashboard layout. 34. The system publishes the updated layout, making it visible to all dashboard viewers.
Alternative flow 1: 6. If the operational manager requires a specific visualisation change, they raise the request to the data analyst who implements and validates the change before publishing.

Acceptance Criteria

19. The edited dashboard layout is saved and correctly displayed to all users after publishing.
20. Added elements display data from the connected data warehouse correctly.
21. Removed elements no longer appear in the published dashboard view.
22. The updated design must be validated and approved by the operational manager before going live.

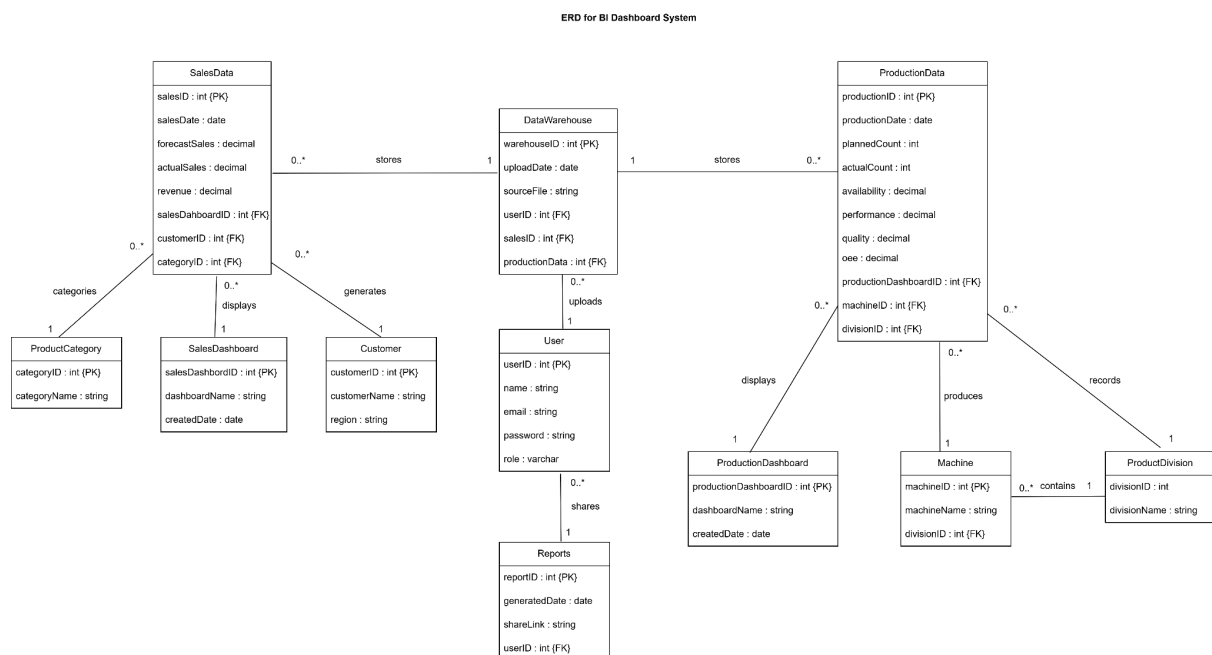
Exception flow 1:

6. If a newly added chart cannot connect to the data source, the system displays a data source error and the analyst must verify the warehouse connection before saving.

Table 1.3.6 User story: Manage dashboard layout

2.1 Entity Relationship Diagram (ERD)

Figure 1 shows ERD for BI Dashboard System. There are 11 main entities in the ERD: User, Data Warehouse, Sales Dashboard, Production Dashboard, Customer, Product Category, Product Division, Machine, Sales Data, Production Data and Reports. SalesData contains the performance of the sales, linked to Customer and ProductCategory. While ProductionData contains the performance of the production, linked to Machine and ProductDivision. The DataWarehouse entity is the one that stores sales and production data uploaded by users, centralizing all data. The SalesDashboard and ProductionDashboard entities are used to present the data, for business performance monitoring and decision making. Users can create and share reports, based on the data from the dashboard. Relationships and cardinalities are used to describe organization of and connections between data in the BI Dashboard System.



2.2 Use Case Diagram

Figure 2 shows the Use Case Diagram of the BI Dashboard System. The system consists of three actors: Top Management, Operational Manager and Data Analyst. Top Management and Operational Managers use the dashboards to monitor sales and production performance, apply filters, perform drill-down analysis and share reports. Data Analysts manage the data warehouse, perform ETL processes and maintain dashboard layouts.



2.3 Data Architecture

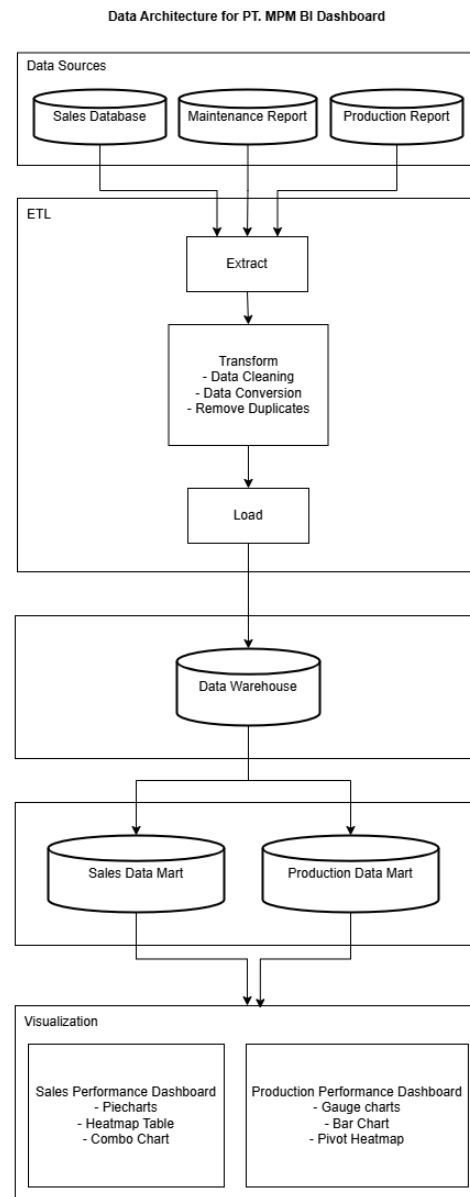


Figure 3: Data Architecture for PT. MPM BI Dashboard

Figure 3 illustrates the data architecture designed to power the PT. MPM Business Intelligence system, demonstrating how raw enterprise data is structured and processed for organizational use. Data Sources serves as the origin layer, encompassing raw sales databases, maintenance logs, and production records generated by core business operations. In ETL Process, the Data Analyst extracts this raw data and performs systematic cleaning, duplicate removal, and data type conversion to ensure high data quality. Once transformed by the ETL pipeline, the standardized data is compiled into a centralized enterprise Data Warehouse hosted on Google Sheets. This layer acts as the single source of truth, housing the consolidated history of all corporate records before they are broken down for departmental access. To optimize analytical performance and isolate business domains, the central data

warehouse partitions its records into two distinct, subject-oriented repositories. The first is Sales Data Mart that is dedicated exclusively to storing processed data structures required for financial tracking, forecast calculations, and revenue analysis. The second is Production Data Mart that is dedicated specifically to handling manufacturing variables, operational availability records, and equipment efficiency logs. The final layer delivers interactive Sales and Production Performance Dashboards containing custom gauge charts, heatmaps, and combo charts directly to Top Management and Operations Managers to support strategic decision-making.

2.4 Multidimensional Model

Figure 4 and Figure 5 outlines the multidimensional modeling strategy applied to the data warehouse using a Star Schema approach to optimize query speeds and analytical slicing within Looker Studio. The architecture is split into two specialized analytical frameworks: the Sales Star Schema and the Production Star Schema. The Sales Star Schema centers around a core Sales Fact Table that stores measurable performance metrics such as revenue, sales volumes, and target forecasts. This fact table links via foreign keys to surrounding dimension tables including Time, Customer, and Product Category Dimensions allowing users to seamlessly filter and drill down into revenue streams by date or client type. Similarly, the Production Star Schema features a central Production Fact Table capturing critical manufacturing variables like planned counts, actual counts, and Overall Equipment Effectiveness (OEE) metrics. This central table is connected to dedicated Machine, Product Division, and Time Dimensions, allowing management to easily identify performance bottlenecks, evaluate machine availability, and measure product quality across specific time periods.

2.4.1 Sales star schema

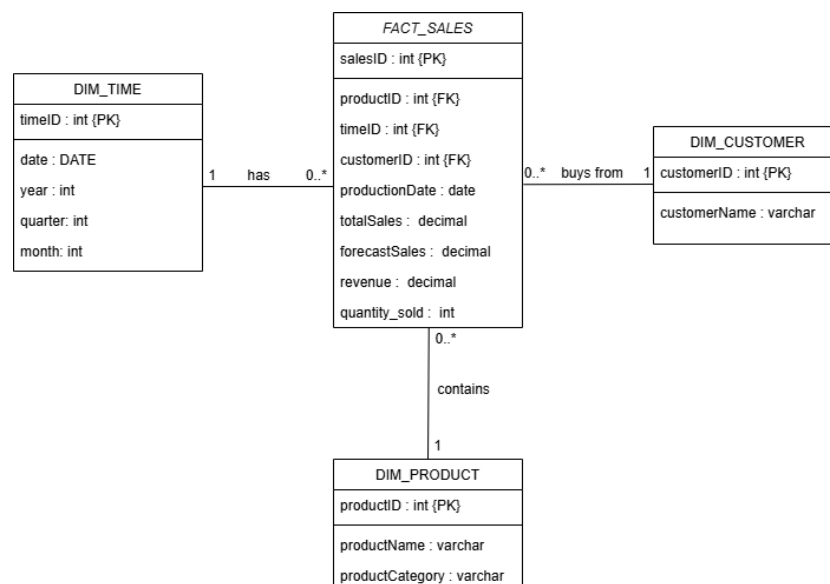


Figure 4: Multidimensional Sales Star Schema Diagram

2.4.2 Production star schema

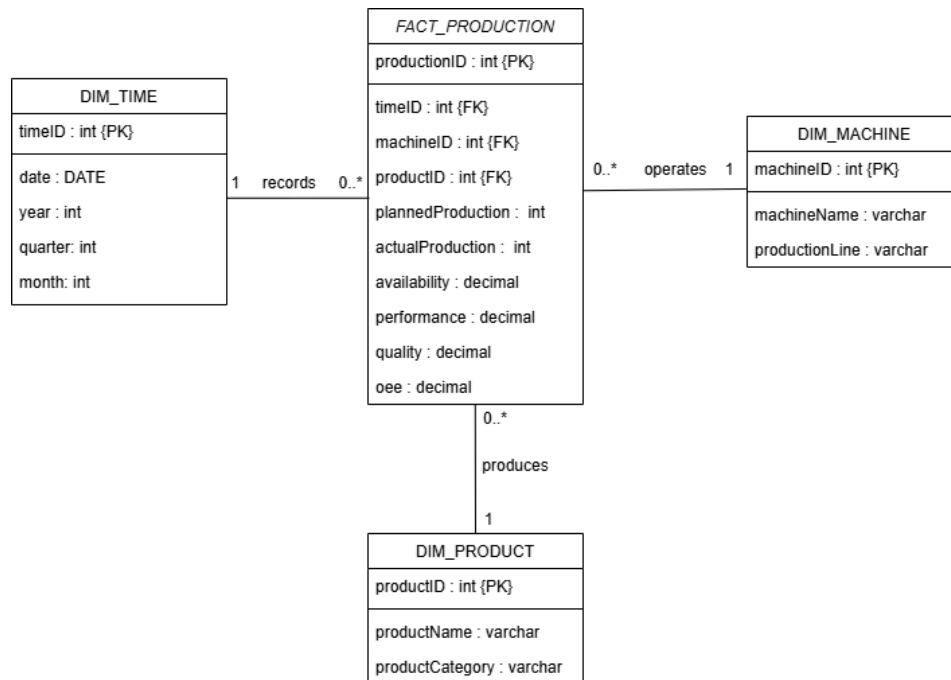


Figure 5: Multidimensional Production Star Schema Diagram

To effectively monitor manufacturing efficiency, the central Production Fact Table explicitly breaks down Overall Equipment Effectiveness (OEE) into its three standard quantitative pillars, which are calculated using the metrics loaded from the data warehouse. First, Availability measures the ratio of actual production time against scheduled operating time, highlighting downtime losses. Second, Performance tracks operational speed by comparing the actual product count against the theoretical maximum planned count over that timeframe. Third, Quality measures the yield of fully functional packages by identifying the percentage of items that pass inspection without defects. By multiplying these three distinct ratios ($\text{Availability} \times \text{Performance} \times \text{Quality}$), the system calculates the final composite OEE percentage, which Looker Studio renders into visual gauge charts to let operational managers instantly identify whether a line is experiencing an availability bottleneck, a machine slowdown, or a drop in quality control.